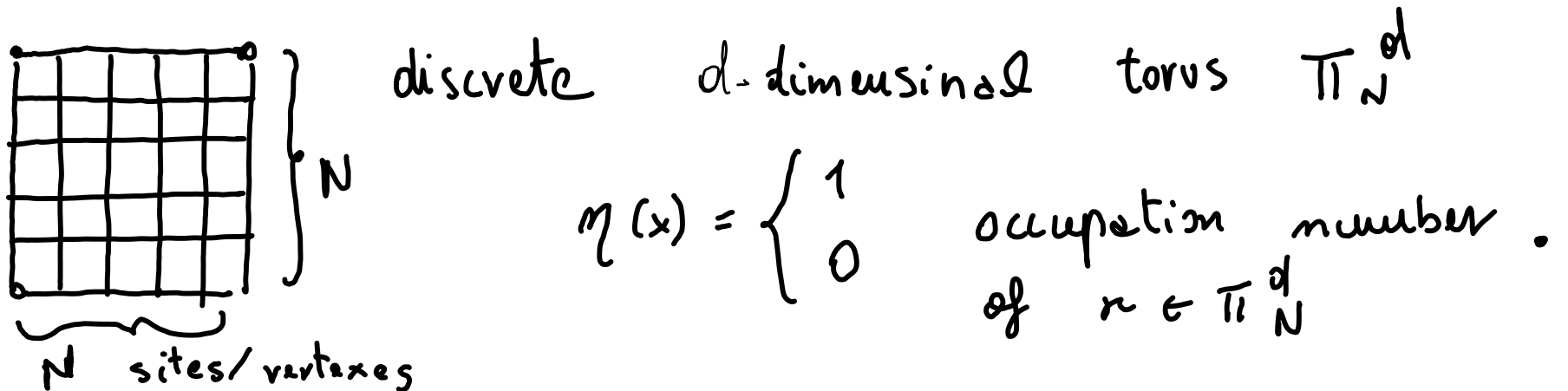
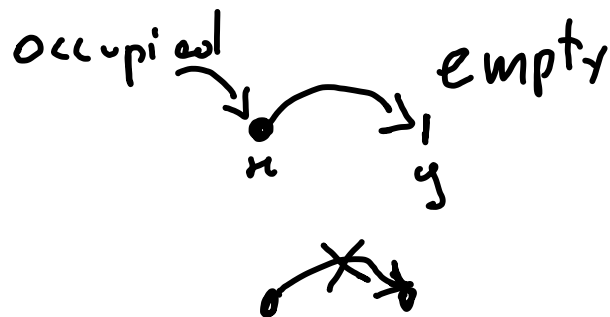


HYDRODYNAMICS OF DIFFUSIVE IPS

Lecture 1: Ch. 4 KL, Hydrodynamics of symmetric ⁽¹⁾ simple exclusion process (SSEP).



configuration $\underline{\eta} := \{\eta(x)\}_x \in \Sigma_N = \{0, 1\}^{\mathbb{T}_N^d}$ state space.



a particle can jump from x to y
if x is occupied and
can not if both are occupied.

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the stochastic dynamics according to a CTMC, defined by the Markov generator $\mathcal{L}_N$: (2)

$$(\mathcal{L}_N f)(\eta) = \sum_{(n,y) \in E_N} \overbrace{\eta(n)(1-\eta(y))}^{\text{exclusion rule}} [f(\eta^{n,y}) - f(\eta)],$$

$f: \Sigma_N \rightarrow \mathbb{R}$, $E_N: \{\text{set of oriented bonds/edges}\}$

$$\eta^{x,y}(z) := \begin{cases} \eta(x)-1 & , z=x \\ \eta(y)+1 & , z=y \\ \eta(z) & , z \neq x,y \end{cases} \quad \begin{array}{l} \text{new configuration} \\ \text{after a jump on } (x,y). \end{array}$$

Fact: $\mathcal{L}_N$ is Bernoulli reversible $\mu_N^p(\eta) = \prod_x \eta^{E_x} (1-\eta(x))^{(1-p)}$
 constant profile.

We want to derive the macroscopic behaviour of observables.

$\underline{\eta}_t := \{\eta_t(n)\}_n$ configuration at time t .

Def. (empirical measure (represents density of particles))

$$\pi_t^N(du, \eta) := \frac{1}{N^d} \sum_{n \in \pi_N^d} \eta_t(n) \delta_{n/N}(du)$$

(3)

$\mathcal{M}_+(\pi^d)$ space of positive measures on $\pi^d := [0, 1]^d$
 $\uparrow$ continuous torus

Remark two space scales:

- the microscopic one: π_N^d (edges of size 1)
- the macroscopic one: π^d

$$x \in \pi_N^d \mapsto x/N \in \pi^d, \quad u \in \pi^d \mapsto [Nu] \in \pi_N^d$$

$$\int_0^1 \pi_t^N(du, \eta) f(u) = \frac{1}{N^d} \sum_{x \in \pi_N^d} f(x) \eta_t(x)$$

• Call $\mathcal{D}([0, T], \Sigma_N)$ space of trajectories of the process $\{\eta_t\}_t$ and $\mathcal{C}([0, T], \mathcal{M}_+(\pi^d))$ that of $\pi_t(m, du)$. CAD LAG — (4)

• $\eta_{t=0}(n)$ are distributed along an initial distribution μ_N .

• Remark We consider the accelerated process

$\eta_{t(N^2)} \leftarrow$ diffusive time scale or $N^2 L_N$.
 $\hookrightarrow$ Two time scale : $\begin{cases} \text{macroscopic } [0, t] \\ \text{microscopic } [0, N^2 t] \end{cases}$

Def Let $\{P_N\}_N$ and $\{Q_N\}_N$ respectively the sequences of probability measures induced by $\{\eta_{N^2 t}\}_t$ on $\mathcal{D}([0, T], \Sigma_N)$ and by $\pi_t^N(du, n)$ on $\mathcal{C}([0, T], \mathcal{M}_+(\pi^d))$.

Statement of Hydrodynamics

(5)

$\rho_0 : \mathbb{T}^d \rightarrow [0, 1]$ initial profile (smooth)

$\{\mu^N\}_N$ the associated sequence of Bernoulli initial measures, i.e. $\mu_N(\eta) = \prod_{x \in \mathbb{T}^d_N} [\eta(x)]^{\rho_0(x/N)} [1 - \eta(x)]^{(1 - \rho_0(x/N))}$.

Then $\forall t \geq 0$, the sequence of random measures $\pi_t^N(\eta)$ converges in Probability to the absolute continuous measures $\rho(t, u) du$, i.e. .

$$\lim_{N \rightarrow \infty} \mathbb{Q}_N \left(\left| \int_{\mathbb{T}^d} f d\pi_t^N(\eta) - \int_{\mathbb{T}^d} f \rho(u, t) du \right| > \varepsilon \right) = 0,$$

$\forall \varepsilon > 0, f \in C(\mathbb{T}^d)$, where $\rho(t, u)$ is the solution (strong) of the Cauchy problem $\begin{cases} \partial_t \rho = \Delta \rho \\ \rho(0) = \rho_0 \end{cases}$.

$$\leadsto \left[\pi_t^N(\eta, du) \xrightarrow[N]{(w)} \rho(u, t) \right] \text{ as } \underline{LLN}$$

5 Steps of the proof

⑥

- 1) Proving that $\{Q_N\}_N$ is relatively compact [Functional Prohorov]
- 2) Identifying the limit points and proving that all converging subsequences $\{Q_{N_k}\}_k$ have the same limit.

Remarks

A) Point 1) requires to set a topology on $\mathcal{C}([0, T], \mathcal{H}^+)$ to characterize compact set

B) Point 2) will be given by the typical behaviour [in probability (LLN)] of $\pi_t^N(\eta, du)$ for $N \gg 1$, i.e. the PDE characterizing the hydrodynamics.

Topology and Compactness

$\mathcal{M}^+(\pi^d)$ is sep. complet metric space with the metric induced by the weak top.

Billingsly, "Convergence of Probability measures", Ch 3, §.12 (7)

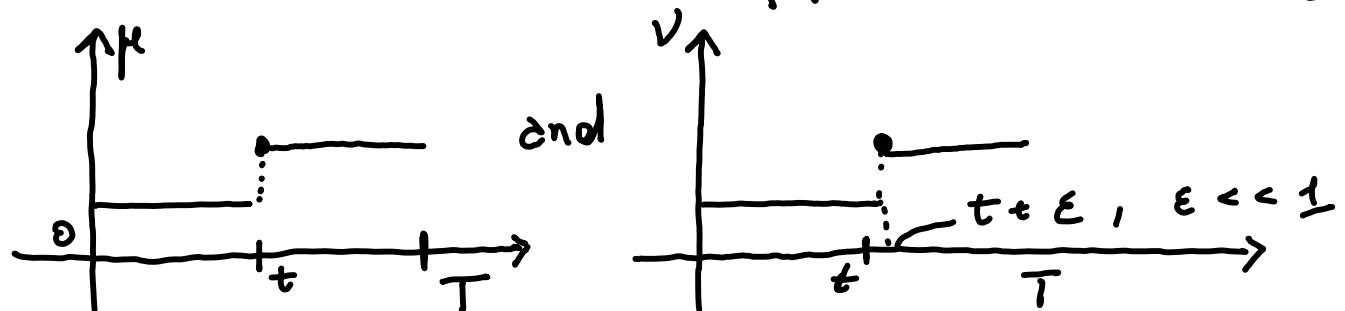
$$g(\mu, \nu) := \sum_{k=1}^{\infty} \frac{1}{2^k} \frac{|\langle \mu, f_k \rangle - \langle \nu, f_k \rangle|}{1 + |\langle \mu, f_k \rangle - \langle \nu, f_k \rangle|}, \quad \{f_k\}_k \text{ dense countable of } C(\pi^d)$$

$$\langle f, \mu \rangle := \int_{\pi^d} f \, d\mu.$$

Remark On $\mathcal{C}([0, T], \mathcal{M}^+)$ the uniform topology

$d(\mu, \nu) = \sup_{s, t \in [0, T]} g(\mu(s), \nu(t))$ is not good because

it evaluates as "very different" measures that differ for a small time difference in a jump, like



$\leadsto$ So it is introduced the Skorohod Topology: 8
 two trajectories $\mu(s)$ and $\nu(s)$ are "close" if deforming
 a little bit of the same time their "ordinate/graph"
 and "time scale" they are "similar".

$$d(\mu, \nu) := \inf_{\lambda \in \Lambda} \max \left\{ \sup_{0 \leq t \leq T} |\lambda(t) - t|, \sup_{0 \leq t \leq T} \rho(\mu(t), \nu(\lambda(t))) \right\}$$

where $\Lambda := \left\{ \begin{array}{l} \text{strictly increasing functions } \lambda: [0, T] \rightarrow [0, T] \\ \text{with } \lambda(0) = 0 \text{ and } \lambda(T) = T \end{array} \right\}$

$\leadsto \mathcal{C}$ becomes sep. complete[⊗] metric space.

[⊗] for real for completeness something slightly different has
 to be considered,

Functional Ascoli-Arzelà: Given $A \subset \mathcal{C}([0,1], \mathcal{M}^+)$ if ⑧

1) $\{\mu_t; \mu \in A, t \in [0,1]\}$ is relatively compact on $\mathcal{M}^+$

[boundedness property, $\mu \in \mathcal{M}^+$ is rel. comp. iff
 $\sup_{\mu \in A} \int_{\mathbb{T}^d} d\mu(x) < +\infty$, i.e. total mass is finite.]
 (total variation norm $\|\mu\|$ of μ .)

2) $\lim_{\delta \rightarrow 0} \sup_{\mu \in A} w_\mu(\delta) = 0$, $w_\mu(\delta) := \sup_{|t-s| \leq \delta} g(\mu_s, \mu_t)$
 [modulus of continuity]

[uniform equicontinuity property, μ is uniformly
 continuous iff $\lim_{\delta \rightarrow 0} w_\mu(\delta) = 0$.]

$\Rightarrow A$ is relatively compact.

Remark: We have to reduce this in probabilistic terms $\leadsto$ Prokhorov's theorem for $\{\mathbb{Q}_N\}_N$.

Reminder (Prohorov's theorem):

Let X compl. sep. metric space and $\{P_N\}_N$ seq. of prob. on it. (10)

$\{P_N\}_N$ is tight iff $\{P_N\}_N$ relatively compact

$$[\forall \varepsilon > 0 \exists K_\varepsilon \mid P_N(K_\varepsilon) \geq 1 - \varepsilon \quad \forall N]$$

$$[\exists \{P_{N_k}\}_{N_k} \mid P_{N_k} \xrightarrow{(w)} P]$$

Functional Prohorov (to prove point 1) of Hydro.)

Let $\{Q_N\}_N$ sequence of pr. meas. on $\mathcal{C}([0, T], \mathcal{H}^+)$ if

1) $\forall t \in [0, T]$ and $\forall \varepsilon > 0$, there is a compact $K(t, \varepsilon) \subset \mathcal{H}_+$ s.t. $\sup_N Q_N[\mu: \mu \notin K(t, \varepsilon)] \leq \varepsilon$
(i.e. we have a tightness of $\{Q_N\}_N$)

2) $\forall \varepsilon > 0$, $\lim_{\delta \rightarrow 0} \limsup_{N \rightarrow \infty} Q_N[\mu: \nu_\mu(\delta) > \varepsilon] = 0$,
(i.e. equicontinuity in probability) $\uparrow$ (all trajectory $\{\mu_t\}_t$)

$\Rightarrow \{Q_N\}_N$ is relatively compact.

Concrete meaning: what to do

(11)

For 1) we show boundedness of the total variation norm on all time

$$\text{i.e. } \lim_{\ell \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathbb{P}_N \left\{ \sup_{0 \leq t \leq T} \frac{1}{N} \sum_x |\eta_t(x)| \geq \ell \right\} \underset{\uparrow}{=} 0$$

[deterministic for SEP]

For 2) we show uniform equicontinuity for each test function $f \in C(\mathbb{T}^d)$

$$\text{i.e. } \forall \varepsilon > 0 \lim_{\delta \downarrow 0} \limsup_{N \rightarrow \infty} \mathbb{P}_N \left\{ \sup_{\substack{0 \leq s, t \leq T \\ |t-s| \leq \delta}} |\langle f, \pi_t^N \rangle - \langle f, \pi_s^N \rangle| \geq \varepsilon \right\} = 0$$

$\{f_k\}$ countable dense set of $C(\mathbb{T}^d)$.

1) is granted for SSEP

(12)

2) Consider $C^2(\pi^d)$ dense set and the following martingale

$$t > s > 0, M(t-s) = \langle \pi_t^N, f \rangle - \langle \pi_s^N, f \rangle - N^2 \int_s^t dr \mathcal{L}_N \langle \pi_r^N, f \rangle \left[\text{Dynkin's formula} \right]$$

$$\mathbb{Q}_N \left(\sup_{\substack{0 \leq s, t \leq T \\ |t-s| < \delta}} |\langle \pi_t^N, f \rangle - \langle \pi_s^N, f \rangle| > \epsilon \right)$$

from Dynkin

$$\leq \mathbb{Q}_N \left(\sup_{\substack{0 \leq s, t \leq T \\ |t-s| < \delta}} |M(t,s)| + N^2 \left| \int_s^t \mathcal{L}_N \langle \pi_r^N, f \rangle \right| > \epsilon \right)$$

$$= \frac{1}{2} \int_s^t dr \langle \pi_r^N, \underbrace{\Delta_N f}_{\substack{\text{discrete} \\ \text{of } f}} \rangle$$

leapfrog

$$\bullet \left| \frac{N^2}{2} \int_s^t dr \langle \pi_r^N, \Delta_N f \rangle \right| \leq \underbrace{c(f)}_{\text{constant}} |t-s|$$

For a CTMC given the martingale

$$M(t) = F(\eta_t) - F(\eta_0) - \int_0^t dr \mathcal{L} F(\eta_r)$$

(13)

$$N(t) = M^2(t) - \underbrace{\int_0^t dr (\mathcal{L} F^2(\eta_r) - 2 F(\eta_r) \mathcal{L} F(\eta_r))}_{\mathcal{L} := B(s)}$$

is also a martingale.

Important trick:

$$(\mathcal{L} F^2 - 2 F \mathcal{L} F)(\eta) = \sum_{\eta'} \underbrace{c(\eta, \eta')}_{\substack{\text{take } F(\eta_r) = \langle \pi_r^N(\eta), f \rangle}} (F_{\eta'} - F_{\eta})^2.$$

$$\text{take } F(\eta_r) = \langle \pi_r^N(\eta), f \rangle$$

$$\mathbb{E}_{\eta_N} (M^2(t-s)) \underset{\substack{\uparrow \\ N(t) \text{ is martingale with initial value zero}}}{=} \mathbb{E}_{\eta_N} \left(\int_s^t B(s) ds \right) \leq c(f) \frac{|t-s|}{N^d}$$

$N(t)$ is martingale with initial value zero

$$Q_N \left(\sup_{\substack{0 \leq t, s \leq T \\ |t-s| > \delta}} |H(t-s)| + \left| N^2 \int_s^t L_N \langle \pi_r^N(m), f \rangle \right| \geq \varepsilon \right) \quad (14)$$

$$\leq Q_N \left(\sup |H(t-s)| > \frac{\varepsilon}{2} \right) + Q_N \left(\sup \left| N^2 \int_s^t L_N \langle \pi_r^N(m), f \rangle \right| > \frac{\varepsilon}{2} \right)$$

from previous estimates and Chebyshev + Doob's ineq.

We have that $\lim_{\delta \uparrow 0} \limsup_{N \rightarrow \infty} Q \left(\sup_{\substack{0 \leq t, s \leq T \\ |t-s| > \delta}} | \quad + \quad | > \varepsilon \right) = 0.$

Second Step: characterization of limit points (hydrodynamics) (15)

Remark: $\sup_{0 \leq t \leq T} |\langle \pi_t^N, f \rangle| \leq \frac{1}{N^d} \sum_x f(x/N) \approx \int_{\mathbb{T}^d} f(u) du + O\left(\frac{1}{N}\right)$

$Q_{N_k} \xrightarrow{(w)} Q^* \leftarrow$ limit point, $Q^* [\pi: \pi_t(du) = \pi_t(u) du] = 1$
 $\uparrow$ concentrated on abs. c. traj.
 w.r.t. to Lebesgue.

Consider the martingale

$$\tilde{M}(t) = \langle \pi_t^N, f \rangle - \langle \pi_0^N, f \rangle - \int_0^t ds \left[\langle \pi_s^N, \partial_s f \rangle + \mathcal{L}_N \langle \pi_s^N, f \rangle \right]$$

Define $\tilde{N}(t)$ and $\tilde{B}(t)$ as before we have

$$\lim_{N \rightarrow \infty} \sup Q_N (|\tilde{M}(t)| > \varepsilon) = 0$$

Remind $N^2 \mathcal{L}_N \langle \pi_s^N, f \rangle = \frac{N^2}{2} \langle \Delta_N f, \pi_s^N \rangle$
 $\rightarrow$ non trivial hydrodynamics $\frac{N^2}{2} \Delta_N f(x) \approx \frac{\Delta}{2} f(x) + O\left(\frac{1}{N^2}\right)$

- Therefore the RHS is a discrete form of weak heat eq.: 16

$$\tilde{H}(t) = \frac{1}{N^d} \sum_x \eta_t(x) f(x) - \frac{1}{N^d} \sum_x \eta_0(x) f(x) - \int_0^t ds \left[\sum_x \eta_s(x) \partial_s f(x) + \frac{N^2}{2} \sum_x \Delta f(x) \eta_s(x) \right]$$

$N \rightarrow \infty$ becomes a weak heat equation, with the

initial condition
$$\int_{\mathbb{T}^d} \rho(0, x) f(x) = \int_{\mathbb{T}^d} d\mu \rho_0(x) f(x)$$

because from the initial distribution $\mu(\eta) = \prod_x (\eta(x))^{p_0(x)} (1 - \eta(x))^{(1-p_0(x))}$,

$$\lim_{N \rightarrow \infty} \mu_N \left(\left| \int_{\mathbb{T}^d} f d\pi_0^N(\eta) - \int_{\mathbb{T}^d} f \rho_0(x) d\mu \right| > \varepsilon \right) = 0, \forall \varepsilon > 0.$$

- So the limit points Q^* are concentrated on sol. of heat eq., moreover since the sol. is unique (and strong) $\leadsto$ so Q^* is unique, i.e.

$$\underbrace{Q_{N_k}}_{\text{subsequence}} \xrightarrow{\frac{(\cdot)}{N_k}} \int \rho(t, u) du \leadsto \text{Solta measure on the unique sol. of } \begin{cases} \partial_t \rho = \frac{1}{2} \partial_{nn} \rho \\ \rho(0, n) = \rho_0(n) \end{cases}$$

- But weak convergence to a deterministic value implies convergence in probability, i.e. $\pi_N^t(\eta, du) \xrightarrow{\frac{1}{N}} \rho(t, u) du$. (17)

Reminder

Prop. Given a sequence $(X_n, \Omega_n, \mathcal{F}_n, \mathbb{P}_n)$ with (S, d) metric space and $X: \Omega_n \rightarrow S$, s.t. $X_n \xrightarrow{\frac{1}{n}} a \in S$, where $a \in S$ is a deterministic value, then $\mathbb{P}_n(d(X_n, a) < \varepsilon) \xrightarrow{n} 1$.
 Namely, $X_n \xrightarrow{\frac{1}{n}} a$ implies $X_n \xrightarrow{\mathbb{P}} a$.

Proof take $f = \chi_G$ where χ_G is the characteristic function of the open set $G = \{u \in S: d(u, a) < \varepsilon\}$,

$$\mathbb{E}_n(\chi_G(X_n)) \xrightarrow{n} \mathbb{E}(\chi_G(a)) = 1$$

$$\int_{\Omega_n} \chi_G(X_n) d\mathbb{P}_n = \mathbb{P}_n(X_n \in G)$$

OBSERVATION: Write the discrete heat eq.

(18)

$$M(t) = \frac{1}{N^d} \sum_x \eta_t(x) f(x) - \frac{1}{N^d} \sum_x \eta_0(x) f(x) - \int_0^t ds \frac{1}{N} \left[\sum_x \eta_s(x) \partial_s f(x) - N^2 \sum_x \eta_s(x) \Delta_u f(x) \right]$$

$\left\langle \pi_t^N, f \right\rangle \quad \left\langle \pi_0^N, f \right\rangle \quad \left\langle \pi_s^N, \partial_s f \right\rangle \quad \left\langle \pi_s^N, \Delta f \right\rangle$

- It is closed in terms of $\pi_t^N(\eta, du)$, in general it is not true. You can have $\eta_s^2(x)$ in place of $\eta_s(x)$
 - In a more general sense we proved we can replace $\eta_s(x)$ with $\mathbb{E}_{\rho(x,t)}(\eta_s(x))$ where the expectation is w.r.t to Bernoulli of constant parameter $\rho(x,t)$.

the stationary measure
is Bernoulli constant
- So when we will have $\eta_s^2(x)$ we will replace this with $\mathbb{E}_{\rho(x,t)}(\eta_s^2(x)) = \rho^2(x,t)$, assuming the equilibrium is still a Bernoulli(ρ).